Ahmedabad University

CSE 631: Machine Learning

Weekly Report(6)

(Group - 9) Learning Machine

Section 1

Submitted to faculty: Prof. Mehul Raval

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# <u>Weekly Progress Report - 6</u>

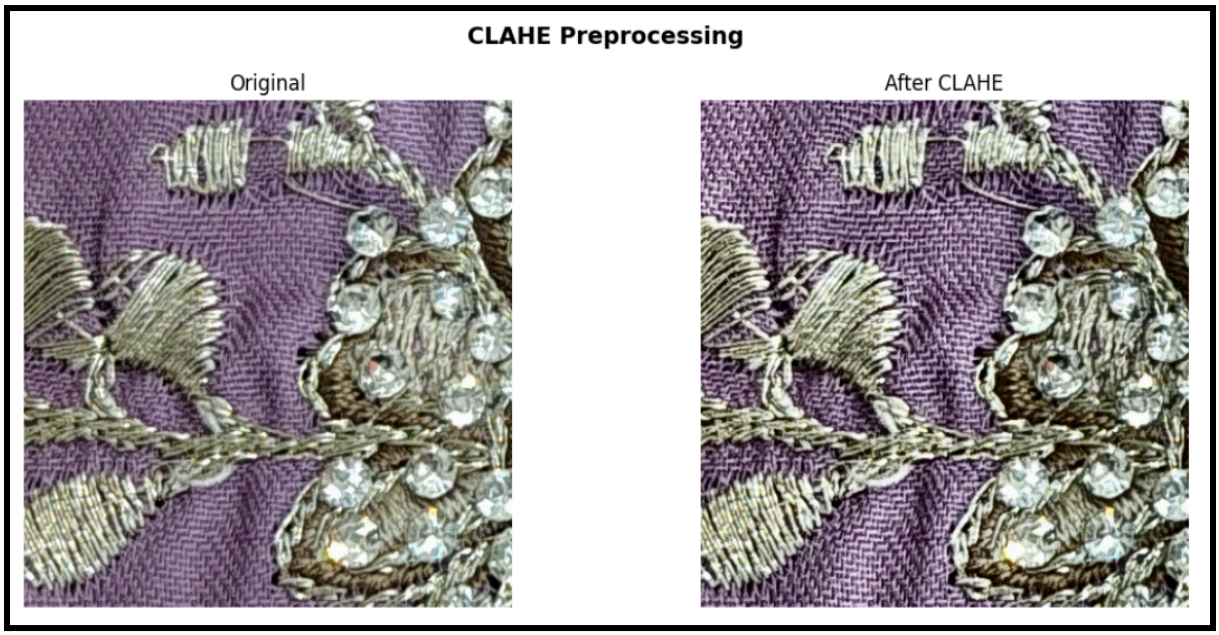
## Machine Learning–Based Automated Stone Counting System for Embroidered Fabrics

1. Activities that were done this week.

The present stage of the project was dedicated to the enhancement of the classical machine learning detection pipeline and its effectiveness by organizing and evaluating experiments.

1.1 Enhancement in Image Preprocessing.

- Preprocessing techniques were optimized so as to enhance the quality of extracting features:
- Used CLAHE (Contrast Limited Adaptive Histogram Equalization) in order to improve image contrast.
- This was highly effective in enhancing features visibility and with ease in classifying them under different lighting environments.
- Preprocessing was found to be essential in improving downstream model.




1.2 Extraction of Features and Model Observations.

- Applied HOG (Histogram of Oriented Gradients) in extraction of features.
- However, it was observed that:
- The HOG characteristics could not be used to incorporate intricate or thick stone designs.
- In crowded or dirty performance was slowed down.
- This pointed out the shortcomings of classical feature descriptors in the specified issue.

1.3 SVM Pipeline Development

- An organized detection chain was developed with the help of SVM (Support Vector Machine):
- SVM kernel Used Gamma non-linear classification.
- Combined 5-fold cross-validation of sound model evaluation and generalization.
- Hyperparameter tuning was done to enhance classification.

1.4 Sliding Window Detection Strategy.

- Applied a sliding window strategy (row scanning) in detecting objects.
- Tuned:
- Window size
- Stride
- This enabled a systematic scan of the whole image in case of any detectors.

1.5 Hard Negative Mining

- Integrated hard negative mining with the aim of enhancing the robustness of the classifier:
- False positives of first detections were gathered.
- These were returned to training as negative samples.
- This helped greatly decrease the cases of false detection later.

Tiles are also used in tile based detection pipeline with a sequence of processing operations applied to each tile.

1.6 Tile-Based Detection Pipeline Tiles Tiles can also be applied using tile based detection pipeline in which a sequence of processing operations are performed on each tile.

- In order to deal with dense areas and big images:
-
- Breaking up of divided images into little tiles.
- Individual tile Applied HOG + SVM detection.
- Integrate the detection of all the tiles into a final result.
- This approach improved:
- Detection coverage
- The management of thick stone formations.

```
Running detection on 28 tiles...

  IMG_20260313_141213615_HDR_tile_0.jpg                    2 stones
  IMG_20260313_141213615_HDR_tile_1.jpg                    0 stones
  IMG_20260313_141213615_HDR_tile_10.jpg                   0 stones
  IMG_20260313_141213615_HDR_tile_11.jpg                   1 stones
  IMG_20260313_141213615_HDR_tile_12.jpg                   25 stones
  IMG_20260313_141213615_HDR_tile_13.jpg                   5 stones
  IMG_20260313_141213615_HDR_tile_14.jpg                   10 stones
  IMG_20260313_141213615_HDR_tile_15.jpg                   2 stones
  IMG_20260313_141213615_HDR_tile_16.jpg                   25 stones
  IMG_20260313_141213615_HDR_tile_17.jpg                   19 stones
  IMG_20260313_141213615_HDR_tile_18.jpg                   22 stones
  IMG_20260313_141213615_HDR_tile_19.jpg                   15 stones
  IMG_20260313_141213615_HDR_tile_2.jpg                    12 stones
  IMG_20260313_141213615_HDR_tile_20.jpg                   43 stones
  IMG_20260313_141213615_HDR_tile_21.jpg                   25 stones
  IMG_20260313_141213615_HDR_tile_22.jpg                   15 stones
  IMG_20260313_141213615_HDR_tile_23.jpg                   4 stones
  IMG_20260313_141213615_HDR_tile_24.jpg                   0 stones
  IMG_20260313_141213615_HDR_tile_25.jpg                   0 stones
  IMG_20260313_141213615_HDR_tile_26.jpg                   0 stones
  IMG_20260313_141213615_HDR_tile_27.jpg                   0 stones
  IMG_20260313_141213615_HDR_tile_3.jpg                    11 stones
  IMG_20260313_141213615_HDR_tile_4.jpg                    28 stones
  IMG_20260313_141213615_HDR_tile_5.jpg                    1 stones
  IMG_20260313_141213615_HDR_tile_6.jpg                    11 stones
  IMG_20260313_141213615_HDR_tile_7.jpg                    7 stones
  IMG_20260313_141213615_HDR_tile_8.jpg                    4 stones
  IMG_20260313_141213615_HDR_tile_9.jpg                    1 stones
```

1.7 Detection Merging and Post-processing.

- Came up with a means of combining multiple tile detections.
- Less duplications are detected and overlapping bounding boxes.
- Increased general chances of detection.

1.8 Model Evaluation

Determined integrity of the pipeline with:

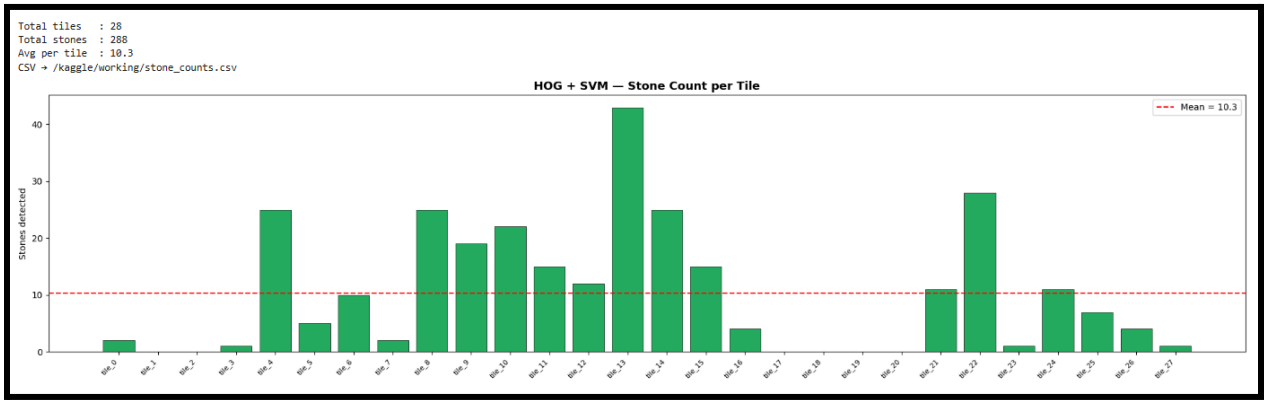
Observed:

- After preprocessing and hard negative mining, the performance is improved.
- Constant restrictions because of HOG feature representation.

```
========================================
  Precision : 0.859
  Recall    : 0.490
  F1 Score  : 0.624
  TP=73  FP=12  FN=76
========================================
                                 tile  GT  Detected  TP  FP  FN
IMG_20260313_141213615_HDR_tile_12.jpg  45        25  20   5  25
IMG_20260313_141213615_HDR_tile_14.jpg  23        10   9   1  14
IMG_20260313_141213615_HDR_tile_16.jpg  35        25  22   3  13
IMG_20260313_141213615_HDR_tile_21.jpg  46        25  22   3  24
```

1.9 Key Findings

- CLAHE also enhanced quality of features and classification.
- HOG + SVM pipeline is functional but has a weakness in the complex and dense environments.
- Hard negative mining was enhanced using tile-based detection.
- Classical ML solution demonstrates evident drawbacks as opposed to projected deep learning functionality.




2. Activities to be done within the next week.

- Additional work on how to make the whole picture correctly out of tile based outputs.
- Proper matching and combination of tiles without any loopholes or deformity.
- tile boundary problem Handle with overlapping tile.
- Enhance detection combination into one final product.
- Increase final visualisation to get clearer results.
- Continue to work on accuracy, precision, recall, and F1-score of the model.
- Reconsider HOG feature extraction, which is not doing wonders, and make a decision about retaining it or changing to a more promising one.